

SWAN HILL, Australia

WMO: 948430

Lat: 35.3767S

Lon: 143.5417E

Elev: 71

StdP: 100.47

Time Zone: 10.00 (AUS)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	0.5	1.6	-7.6	2.0	27.8	-5.2	2.4	25.0	12.5	12.7	11.3	12.6	1.6	160	0.511

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	15.6	39.2	18.6	36.7	18.1	34.6	17.7	21.7	29.5	20.7	29.4	19.8	29.3	5.8	330

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.8	14.7	23.7	18.3	13.3	23.1	16.8	12.0	22.6	63.9	29.0	60.0	28.8	56.8	29.2	34.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.3	9.8	8.6	-1.9	43.0	1.3	2.0	-2.9	44.4	-3.6	45.6	-4.3	46.7	-5.2	48.2		
(5)			-1.4	24.0	1.2	3.0	-2.2	26.2	-2.9	28.0	-3.5	29.7	-4.4	31.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	16.4	24.5	23.9	20.7	16.3	12.6	9.9	9.3	10.2	12.7	15.8	19.4	21.9		
	DBStd	6.37	4.34	4.17	3.91	3.34	2.69	2.27	2.18	2.42	3.22	3.86	4.36	4.23		
	HDD10.0	107	0	0	0	1	6	28	41	26	6	1	0	0		
	HDD18.3	1386	2	3	18	79	179	252	281	253	172	100	36	12		
	CDD10.0	2443	449	391	332	189	86	26	18	32	88	181	283	368		
	CDD18.3	680	193	161	92	18	1	0	0	0	4	22	69	122		
	CDH23.3	9299	2672	2021	1098	243	10	0	0	3	83	403	1019	1747		
	CDH26.7	4704	1519	1071	491	64	0	0	0	0	19	156	485	900		
(14) Wind	WSAvg	4.1	4.5	4.2	4.0	3.5	3.7	3.6	3.9	4.2	4.3	4.4	4.4	4.6		
(15) Precipitation	PrecAvg	337	29	18	24	25	33	27	33	37	36	34	25	19		
	PrecMax	632	111	95	128	84	94	90	99	90	86	116	72	107		
	PrecMin	132	0	0	0	0	1	0	3	5	3	1	2	0		
	PrecStd	112	28	23	28	21	23	19	21	22	22	30	20	21		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	43.0	41.2	37.9	31.5	24.7	19.5	19.0	23.0	29.1	34.6	38.3	40.3		
		MCWB	18.9	19.7	17.8	16.2	13.7	12.6	10.9	12.7	14.1	15.6	17.1	17.9		
	2%	DB	39.9	37.7	34.2	28.3	22.1	17.2	16.7	19.9	25.5	30.8	34.8	37.2		
		MCWB	18.9	19.1	17.8	15.3	13.6	11.5	10.7	11.5	13.2	14.8	16.9	17.5		
	5%	DB	36.9	35.0	31.7	26.1	19.9	15.8	15.1	17.7	22.7	27.8	31.9	34.6		
		MCWB	18.4	18.4	17.3	14.5	12.9	11.2	10.2	10.6	12.6	14.1	15.8	17.2		
	10%	DB	34.2	32.8	29.1	23.8	18.2	14.6	13.8	16.0	20.0	24.9	29.1	31.9		
		MCWB	18.3	18.1	16.8	14.1	12.3	10.6	9.7	10.1	11.9	13.4	15.5	16.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	23.4	23.5	21.0	18.0	16.6	14.5	13.1	14.1	16.5	18.4	20.7	21.8		
		MCDB	28.8	31.0	29.5	23.2	19.1	16.5	16.0	18.9	22.2	26.0	28.5	28.5		
	2%	WB	21.6	22.1	19.8	16.9	14.8	12.8	11.8	12.6	14.9	16.7	19.1	19.9		
		MCDB	29.9	29.8	27.8	23.5	19.1	15.5	14.6	17.4	21.3	24.6	26.9	27.7		
	5%	WB	20.6	20.9	18.8	15.9	13.9	12.0	10.9	11.5	13.7	15.6	17.8	18.7		
		MCDB	30.6	28.8	27.4	23.3	18.3	14.5	13.8	16.4	20.2	23.6	26.8	29.4		
	10%	WB	19.6	19.8	17.7	15.0	13.0	11.2	10.2	10.6	12.6	14.4	16.7	17.6		
		MCDB	30.7	29.3	27.0	22.6	17.3	13.9	13.0	15.1	19.0	23.3	26.1	29.2		
(35) Mean Daily Temperature Range	MDBR		15.6	14.6	13.9	13.5	11.2	9.7	9.6	11.4	13.4	15.0	15.1	15.7		
	5% DB	MCDBR	19.0	17.5	17.1	16.8	13.6	11.3	11.3	14.5	17.3	19.6	18.8	18.8		
		MCWBR	5.6	5.3	5.8	7.0	6.7	6.4	6.3	7.2	7.4	7.3	6.1	5.8		
	5% WB	MCDBR	14.1	12.7	13.4	13.3	10.6	8.9	9.3	12.1	14.0	14.7	13.7	14.0		
		MCWBR	5.4	4.7	5.4	6.1	5.9	5.7	5.8	6.8	7.0	6.6	5.6	5.7		
(40) Clear-Sky Solar Irradiance	taub		0.337	0.337	0.319	0.309	0.298	0.289	0.291	0.299	0.326	0.329	0.343	0.332		
	taud		2.502	2.517	2.556	2.557	2.566	2.590	2.576	2.541	2.452	2.460	2.454	2.501		
	Ebn at Noon		998	977	958	911	865	848	865	908	934	974	986	1008		
	Edh at Noon		113	107	96	84	73	67	71	84	104	112	118	114		
(44) All-Sky Solar Radiation	RadAvg		7.87	7.05	5.77	4.23	2.94	2.35	2.54	3.52	4.84	6.28	7.12	7.89		
	RadStd		0.36	0.33	0.27	0.20	0.15	0.18	0.12	0.22	0.34	0.41	0.38	0.33		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
(46)	Station Only	(d) +0.45	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) +157	(m) +126	(46)
(47)	Regional (12 neighbors)	+0.45	N/A	N/A	N/A	N/A	N/A	N/A	-60	+146	+101	(47)

Nomenclature: See separate page